

Q1: [20] Three identical charges of charge Q are located at the vertices of an equilateral triangle of side length a . Determine the force on one of the charges due to the other two.

Q2: [40] A total charge of Q_1 is distributed uniformly along a semicircular ring as shown in Figure 1. Two point charges, each of magnitude Q_2 , are situated as shown. The surrounding medium is free space.

- (a) Find Q_2 in terms of Q_1 so that the potential Φ at the center of the ring is zero.
- (b) Find Q_2 in terms of Q_1 so that the electric field $\mathbf{E}$ at the center of the ring is zero.

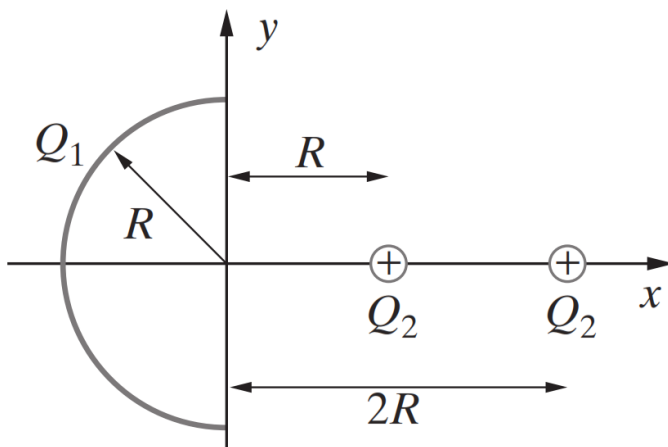


Figure 1: Figure for Question 2.

Q3: [40] The space between two concentric spheres of radii a and b ($a < b$) in free space is charged to a volume charge density given by

$$\rho(r) = \frac{K}{r^2}; \quad a \leq r \leq b$$

where K is a constant. (a) Find the total charge in the shell. (b) Find the electric field at all points in space. (c) Find the electric potential at all points in space. (d) What happens if $b \rightarrow a$?

[25 points] Q4: Applying Coulomb's law, find the electric field at point P with Cartesian coordinates $(0, 0, h)$ due to a circular disk of radius a residing in the x - y plane, and assume that the surface charge density to vary with r as

$$\rho_s = c\rho^3 \quad (c > 0) \quad (\text{C/m}^2)$$

[25 points] Q5: In a certain region of space, the electric field is given in spherical coordinates as follows:

$$\mathbf{E} = \frac{c}{r^4} \mathbf{a}_r \quad (\text{V/m}) \quad (c > 0)$$

Applying Coulomb's law, find the electric potential of point A with respect to point B where A is at $+2$ m and B at -4 m, both on the x -axis.

Hint: Since both field points are on x -axis, therefore, one may write

$$\mathbf{E} = \pm \frac{20}{x^4} \mathbf{a}_x \quad (\text{V/m})$$

[25 points] Q6: In a certain region of space, the volumetric charge density is given in cylindrical coordinates by the function:

$$\rho_v = c \rho e^{-\rho} \quad (c > 0) \quad (\text{C/m}^3)$$

(a) **Apply** the integral form of Gauss's law $\oint_S \mathbf{D} \cdot d\mathbf{s} = Q_{\text{enclosed}}$ to find $\mathbf{D}$.

(b) **Apply** the differential form of Gauss's law $\nabla \cdot \mathbf{D} = \rho_v$ to find $\mathbf{D}$.

Hint: In both cases, one may assume $\mathbf{D} = D_\rho \mathbf{a}_\rho$.

[25 points] Q7: A semi-circular ring of radius a carries a uniform line charge density ρ_L [C/m] and is placed on the xy -plane with axis the same as the z -axis. Obtain the electric field intensity $\mathbf{E}$ at the origin $(0, 0, 0)$ by using the following expression:

$$\mathbf{E}(\mathbf{r}) = \int_L \frac{\rho_L dl' (\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3}$$

and show that

$$\mathbf{E}(0, 0, 0) = -\frac{\rho_L}{2\pi\epsilon_0 a} \mathbf{a}_y$$

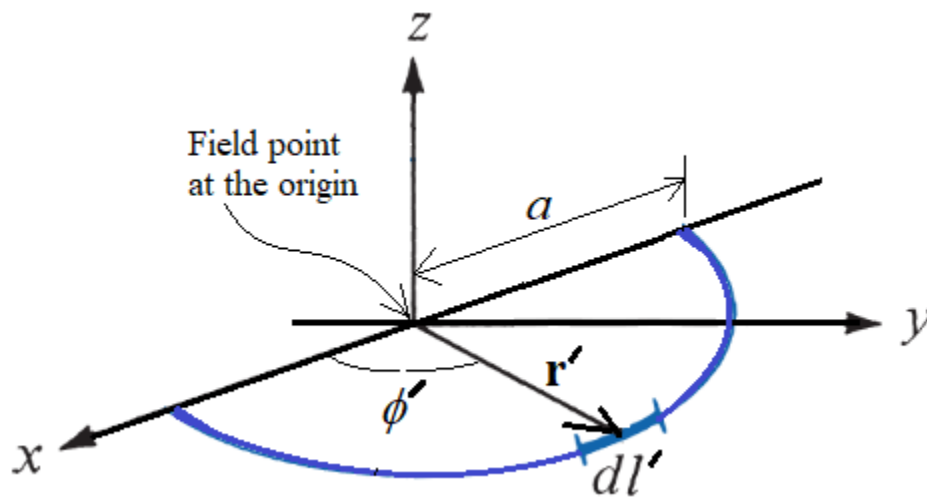


Figure 1: $\mathbf{E}$ field due to a semi-circular ring of charge.

Start with specifying the position vectors $\mathbf{r}$ and $\mathbf{r}'$ for the field point and (differential) source element, respectively, and continue to discuss the detailed evaluation of integral function involved in this problem. Once you have a complete expression of integral function, you may happen to drop one or more terms owing to any particular symmetry in the charge configuration.